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CRSALE010

MAM1019H

Hand-in 6

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Answers

1. (a) False
   (b) False
   (c) False
   (d) False

2.
$$\begin{aligned} n \cdot 3 &= n \cdot s(2) && s(2) = 3 \\ &= n \cdot 2 + n && \text{(M2) with } m = 2 \\ &= n \cdot s(1) + n && s(1) = 2 \\ &= (n \cdot 1 + n) + n && \text{(M2) with } m = 1 \\ &= (n + n) + n && \text{We are given that } n \cdot 1 = n \end{aligned}$$
This means $n \cdot 3 = (n + n) + n$

3.
$$\begin{aligned} 2 \cdot 1 &= 2 \cdot s(0) && s(0) = 1 \\ &= 2 \cdot 0 + 2 && \text{(M2) with } n = 2, m = 0 \\ &= 0 + 2 && 2 \cdot 0 = 0 \text{ by (M1)} \\ &= 0 + s(1) && s(1) = 2 \\ &= s(0 + 1) && \text{(A2) with } n = 0, m = 1 \\ &= s(0 + s(0)) && s(0) = 1 \\ &= s(s(0 + 0)) && \text{(A2) with } m = 0, n = 0 \\ &= s(s(0)) && 0 + 0 = 0 \text{ by (A1)} \\ &= s(1) && s(0) = 1 \\ &= 2 && s(1) = 2 \end{aligned}$$
This means that $2 \cdot 1 = 2$